

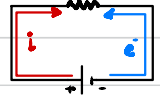
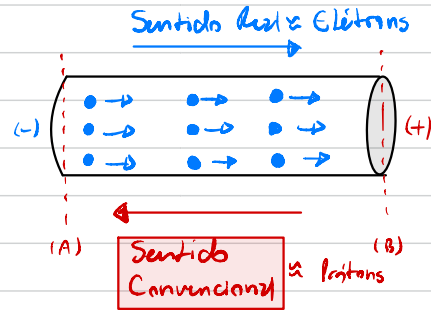
ELETRODINÂMICA

Fluxo

1 Corrente Elétrica (i)

↳ Fluxo ordenado de cargas elétricas

* Só há formação de **corrente** se houver **d.d.p** entre (A) e (B).

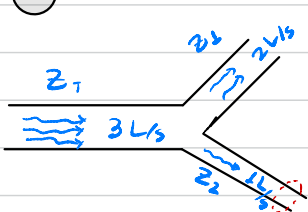


Corrente $\approx$ Fluxo $\approx$ Vazão

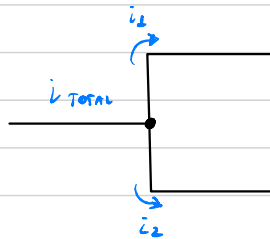
$$i = \frac{\text{Carga (Q)}}{\text{tempo (\Delta t)}}$$

$$Z = \frac{V_{ab}}{\Delta t}$$

I

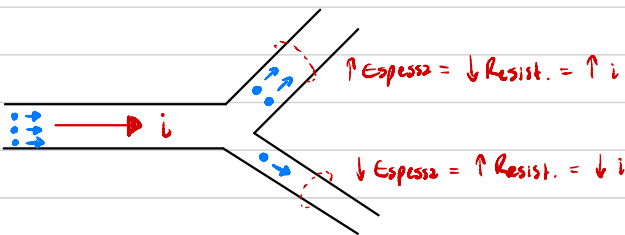


$$Z_{\text{TOTAL}} = Z_1 + Z_2$$



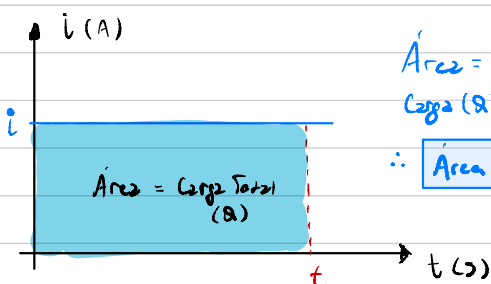
$$i_{\text{TOTAL}} = i_1 + i_2$$

II



1.1 Gráfico (i x t)

$$i = \frac{Q}{\Delta t} \Rightarrow Q = i \cdot \Delta t$$



$$\begin{aligned} \text{Área} &= b \cdot h \\ \text{Carga (Q)} &= i \times t \\ \therefore \text{Área} &\approx Q \end{aligned}$$

• Unid. de Medida

$$[i] = \frac{[Q]}{[t]} = \frac{1 \text{ Coulomb}}{1 \text{ segundo}} = 1 \text{ Ampère}$$

* Baterias eletrônicas:

$$[mAh] = \text{Carga (Q)}$$

$$i = \frac{Q}{t}$$

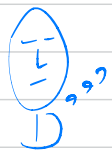
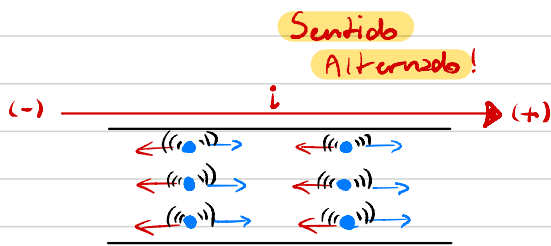
$$\text{Ex: } 4000 \text{ mAh} = 4 \text{ A} \cdot \text{h}$$

P) Carregar essa bateria é necessário $i = 4 \text{ A}$ por 1 h .

$$4 \text{ A} \rightarrow 1 \text{ h} \quad 4 \cdot 1 = 2 \cdot x$$

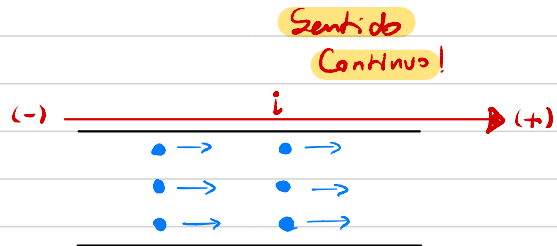
$$2 \text{ A} \rightarrow x \quad x = 2 \text{ h}$$

1.2 Corrente Alternada vs Corrente Contínua



Indução Magnética
(Lei de Faraday-Lenz)

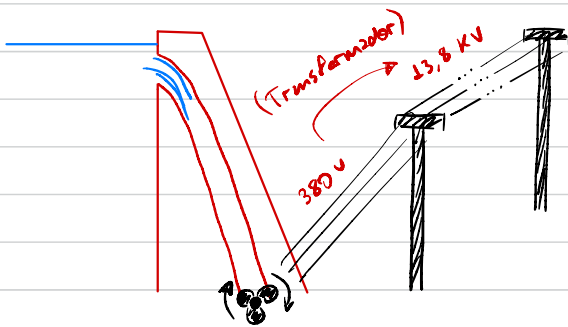
VARIAÇÃO de FLUXO MAG
Gera Corrente Elétrica



"Sensível"

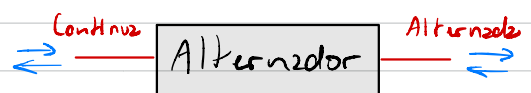
Aplicações:

- 1) Geração/Transmissão de Energia
- 2) Transformadores
- 3) Instalações Elétricas
- 4) Ventiladores, Motores



Aplicações:

- 1) Circuitos Eletrônicos
- 2) Pilhas e Baterias
- 3) Placas Fotovoltaicas
- 4) Circuitos de carros/motos



• Guerra das Correntes:

↳ Nikola Tesla vs Thomas Edison

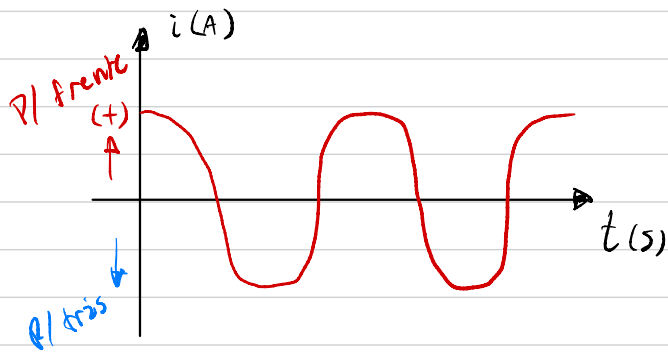


Corrente Alternada

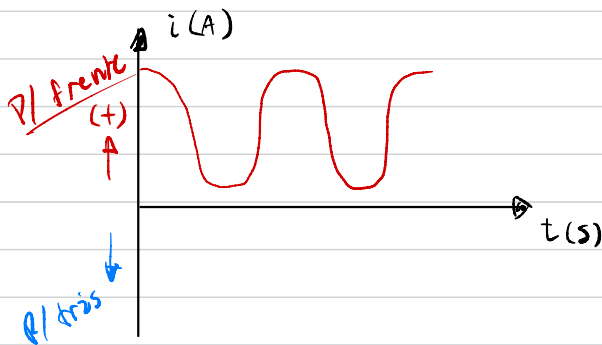
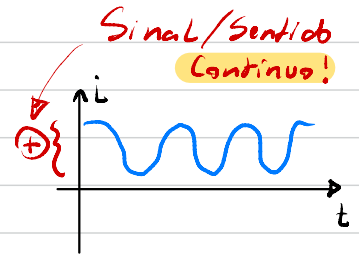


Corrente Contínua

• Gráfico Corrente Alternada vs Contínua



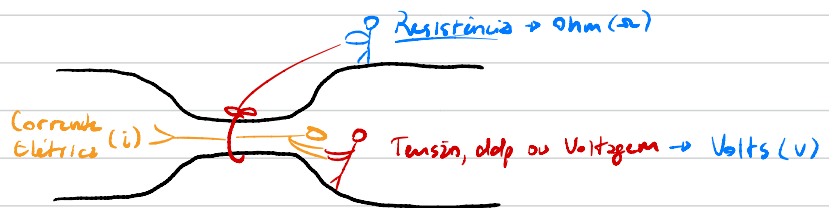
Corrente Alternada!



Corrente Contínua!

② Tensão e Resistência

d.d.p (U)



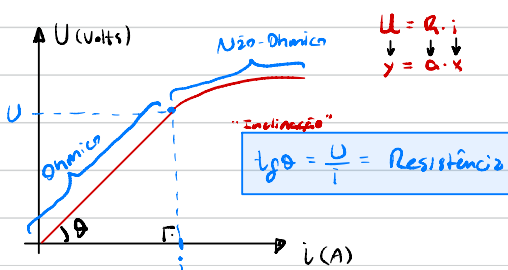
↑ Resistência = ↓ Corrente
↑ Resistência = ↑ Tensão

2.1 1ª Lei de Ohm:

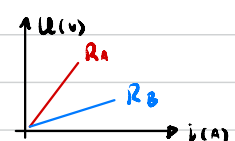
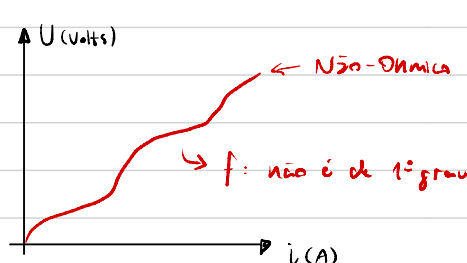
$$\begin{aligned} & \uparrow R = \downarrow i \\ & \uparrow R = \uparrow U \end{aligned} \quad R = \frac{U}{i} \quad \text{ou} \quad U = R \cdot i$$

Quando a Resistência: **R - Ôhmica** é constante

Ⓘ Resistores Ôhmicos

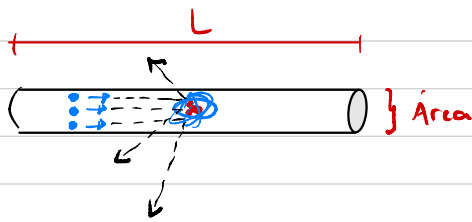


Ⓢ Resistores Não-Ôhmicos

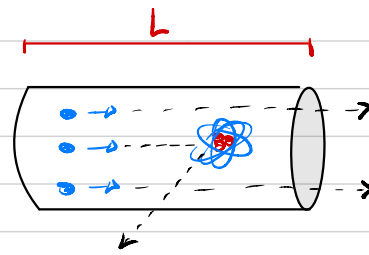


Note que: **R_A > R_B**

2.2 2ª Lei de Ohm $\approx$ Características Geométricas e Materiais



$\uparrow L = \uparrow \text{Resist.}$
 $\downarrow \text{Área} = \uparrow \text{Resist.}$



$\downarrow L = \downarrow \text{Resist.}$
 $\uparrow \text{Área} = \downarrow \text{Resist.}$

$$R = \rho \cdot \frac{L}{A}$$

- L: Comprimento
- A: Área
- ρ : Resistividade $\rightarrow$ Isolantes: $\uparrow \rho$
 $\rightarrow$ Condutores: $\downarrow \rho$

• Aplicações

① Termosensores: $\uparrow \text{Temperatura} = \uparrow \text{Resist.}$

② Super-Condutores: N_2 Líquido $\rightarrow$ Resfriamento $\rightarrow \lll \text{Resistência} = \uparrow \uparrow \uparrow \text{Condutividade} = \text{Super Condutor!}$

Condutividade (σ)

③ Potência e Energia

- Energ. Mecânica
- Energ. Química
- Energ. Nuclear
- Energ. Elétrica

$\uparrow \text{Energia} = \uparrow \text{Pot}$
 $\downarrow \text{tempo} = \uparrow \text{Pot}$

$$P_{\text{ot}} = \frac{\Delta E}{\Delta t} \quad \text{ou} \quad E = P_{\text{ot}} \cdot \Delta t$$

3.1 Potência Elétrica

$$\hookrightarrow P_{\text{ot}} = \frac{E_p}{\Delta t} \quad \text{com } E_p = q \cdot U \quad = \frac{q \cdot U}{\Delta t} \quad \therefore \quad P_{\text{ot}} = i \cdot U \quad \text{"Piu"}$$

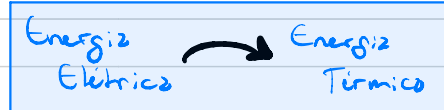
Corrente Elétrica (i)

3.2 Potência e Resistência

$$\begin{array}{l}
 \boxed{P = i \cdot U} \xrightarrow{(U = R \cdot i)} P = i \cdot (R \cdot i) \longrightarrow \boxed{P = R \cdot i^2} \\
 \boxed{P = i \cdot U} \xrightarrow{(i = \frac{U}{R})} P = \left(\frac{U}{R}\right) \cdot U \longrightarrow \boxed{P = \frac{U^2}{R}}
 \end{array}$$

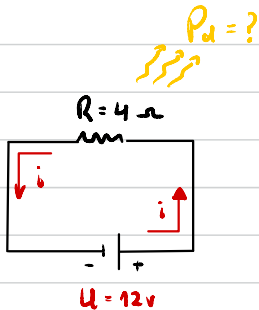
• Obs:

↳ Quando a Pot. está relacionada aos Resistores, geralmente nomeamos de Pot dissipada!



↳ "Dissipador" de Energia

• Ex:



① $i = ?$

$$U = R \cdot i$$

$$12 = 4 \cdot i$$

$$i = \frac{12}{4} \therefore \boxed{i = 3A}$$

② $P_d = ?$

$$1) P_d = \frac{U^2}{R} = \frac{12^2}{4} = \frac{144}{4} \therefore \boxed{P_d = 36W}$$

$$2) P_d = i \cdot U = 3A \cdot 12V \therefore \boxed{P_d = 36W}$$

$$3) P_d = R \cdot i^2 = 4 \cdot 3^2 \therefore \boxed{P_d = 36W}$$

• Resumo de fórmulas (Eletrodinâmica)

$i = \frac{q}{\Delta t}$		
$U = R \cdot i$	$R = \rho \cdot \frac{L}{A}$	
$P = i \cdot U$	$P = R \cdot i^2$	$P = \frac{U^2}{R}$